

Summit Owners Manual (2024)

Introduction and model identification

This manual covers the 2024 THOR Summit 32 and Summit 36 Class A diesel motorhomes. Both are built on the same rear-engine chassis and share the house systems described here. They differ in length, wheelbase, tank capacities, and weight ratings. Keep this manual in the coach and bring it to every dealer visit.

The VIN is on the chassis certification label in the driver-side door jamb and on a plate at the base of the windshield. The first seven characters identify the model:

Model	VIN prefix	Length	Wheelbase	Dry weight	GVWR	GCWR
Summit 32	1THRS32	33 ft 2 in	208 in	22,400 lb	26,000 lb	36,000 lb
Summit 36	1THRS36	37 ft 6 in	242 in	25,100 lb	30,000 lb	40,000 lb

The eighth character is the model year code (R for 2024). The build date is printed on the label as MM/YYYY and is used by service bulletins to define affected units. Maximum towed vehicle weight is 10,000 lb on either model using the 2 inch receiver, which is rated for 1,000 lb tongue load.

Warranty

THOR warrants the Summit house and its THOR-installed systems against defects in materials and workmanship for 36 months from the date of delivery to the first retail owner. The chassis, engine, transmission, and drive axle are warranted separately by the chassis manufacturer for 60 months or 100,000 miles, and those claims go to a chassis service center, not a THOR dealer.

What is covered

The 36 month THOR warranty covers the coach structure, roof, sidewalls, slide-out mechanisms, hydraulic leveling system, generator installation, house electrical, plumbing, appliances, entry step, and cabinetry. THOR handles appliance warranty through its dealers; the owner does not contact appliance makers directly.

What is excluded

Tires, chassis batteries, wiper blades, filters, fuses, and bulbs are excluded. House batteries (SKU BAT-AGM-100) are covered for 12 months. Sealant is a maintenance item; water damage from neglected sealant is not covered. Damage from overloading, driving with slides or jacks extended, or unapproved modifications is excluded.

Claim process

Bring the coach to an authorized THOR dealer. The dealer records the VIN, odometer, and delivery date and submits the claim through the dealer portal. Repairs above 500 dollars need THOR pre-authorization before work begins. Claims must be submitted within 30 days of repair completion and are paid within 45 days of approval.

Chassis and engine

The Summit uses a 6.7 liter inline six-cylinder turbo diesel rated at 340 horsepower and 700 lb-ft of torque, paired with a six-speed automatic transmission. The engine requires ultra-low-sulfur diesel and diesel exhaust fluid (DEF). The DEF tank holds 8 gallons and is filled through the blue cap next to the fuel filler on the driver side. The engine will derate to 5 mph if DEF runs out; a low-DEF warning appears at 15 percent remaining.

Fuel capacity is 100 gallons on the Summit 32 and 120 gallons on the Summit 36. Expect 8 to 9 miles per gallon unloaded and 6 to 7 while towing.

Engine oil capacity is 28 quarts with filter. Change oil and filter every 12 months or 15,000 miles. The fuel filter is changed at the same interval. The air filter has a restriction indicator; replace when the indicator shows red. Tires are 275/80R22.5 and run at 110 psi front and 100 psi rear cold on the Summit 32; the Summit 36 runs 115 psi front and 105 psi rear. Torque lug nuts to 450 ft-lb.

The exhaust brake is engaged by the dash switch and activates whenever the throttle is released. Use it on every downgrade to save the service brakes. The air brake system needs 100 psi before the parking brake will release; the low-air alarm sounds below 60 psi.

Leveling jacks

The Summit has four hydraulic leveling jacks, SKU JACK-HYD-L, powered by a pump under the entry step. Each jack has a 10 inch stroke and lifts 8,000 lb. The control panel is at the driver seat.

Operation

Set the parking brake and put the transmission in neutral. Press AUTO LEVEL and the system extends the jacks to level the coach front to back and side to side within 0.5 degrees. Manual mode extends individual jacks. Never lift a wheel fully off the ground; the coach should remain on its tires. Press RETRACT before driving. The dash chime sounds if the parking brake is released with any jack extended.

Fault lights

Light	Meaning	Action
JACK DOWN (red, steady)	One or more jacks not fully retracted	Retract; check each jack visually
LOW VOLTAGE (amber)	Pump supply below 11 volts	Start engine or connect shore power
EXCESS SLOPE (amber)	Site slope exceeds 6 degrees	Move the coach to flatter ground
LOW FLUID (red)	Reservoir below minimum	Check for leaks; fill with hydraulic fluid
SENSOR (red, flashing)	Level sensor not communicating	Cycle 12V power; dealer to check harness

Bleed procedure

Air in the system causes jacks to creep down overnight. To bleed: extend all jacks fully with the pump running until each cylinder reaches the end of stroke, hold for 10 seconds, then retract fully. Repeat three times. Check the reservoir; it should be within 1 inch of the top with all jacks retracted. Use only the hydraulic fluid specified on the reservoir label. If a single jack still creeps more than 1/4 inch in 8 hours after bleeding, its cylinder seal has failed and the jack should be replaced.

Generator

The generator is an 8.0 kW diesel unit in the front cap, fed from the chassis fuel tank. It will not start if the fuel level is below 1/4 tank; this protects the coach from being stranded. Start it from the panel in the galley or the remote at the dash. Warm the generator for 2 minutes before applying load and run it under load for 30 minutes each month during storage. Change the generator oil every 150 hours; capacity is 4 quarts. The generator powers both roof air conditioners, the water heater element, and the converter at the same time. It shuts down on low oil pressure, high coolant temperature, or overspeed and shows a two-digit fault code on the start panel. Code 14 is high temperature, code 27 is low oil, code 36 is overcrank.

Slide-outs

The Summit has two slide-outs: a full-wall living room slide on the driver side and a bedroom slide on the passenger side. Both are hydraulic and share the leveling jack pump. The coach must be leveled and the engine off before operating the slides. Hold the switch until the room stops.

The slide seals are dual bulb seals with a wiper. Clean and treat them every 3 months with dry silicone spray. If a slide moves unevenly or one end lags, the hydraulic cylinders are out of sync; run the slide fully in and fully out twice to

resync. If a slide stalls, check the LOW FLUID light, since the slides and jacks share a reservoir. The manual override is a hand pump under the entry step; 60 strokes retracts a slide.

Roof

The roof is a one-piece fiberglass panel with sealed seams along the front cap, rear cap, and around every roof penetration. The seams are sealed with a self-leveling sealant, SKU ROOF-SEAL-Q, sold in 10.3 oz cartridges. One cartridge covers about 12 linear feet of seam.

Inspect all roof seams every 6 months and after any hail event. Look for cracks, lifting edges, or gaps between the sealant and the fiberglass. Clean the area with isopropyl alcohol, remove any loose sealant, and apply ROOF-SEAL-Q over the seam so it extends at least 1/2 inch onto sound material on each side. Do not apply below 50 degrees F. Full cure is 24 hours. Never use silicone sealant on the roof; ROOF-SEAL-Q will not bond to it.

The roof ladder is rated for 250 lb. Walk only on the reinforced area between the roof rails.

Refrigerator

The refrigerator is a 16 cubic foot absorption unit running on propane or 120V. The cooling unit, SKU FRIDGE-CU, is the finned assembly on the back of the cabinet. It contains ammonia under pressure and is not repairable in the field; a yellow residue or ammonia smell at the rear vent means the cooling unit has leaked and must be replaced as a complete assembly.

The refrigerator must be within 3 degrees of level to cool properly when parked. Operating it more than 6 degrees off level for more than 30 minutes can permanently damage the cooling unit. Allow 12 hours to reach temperature. On propane it consumes about 1.5 lb per day. The rear vent must be clear; clean the burner and flue annually.

Battery isolator relay

The Summit has two battery banks: two chassis batteries for starting the engine and four house batteries (SKU BAT-AGM-100) for the coach. A battery isolator relay, SKU ISO-RLY-200, connects the two banks when the engine is running so the alternator charges both, and disconnects them when the engine is off so house loads cannot drain the chassis batteries. It is rated for 200 amps continuous and is mounted in the battery compartment behind the driver-side front wheel.

The relay is also energized by the BATTERY BOOST switch at the dash for 30 seconds at a time, which lets the house batteries help start the engine. A relay that fails closed will drain the chassis batteries overnight through house loads. A relay that fails open means the house batteries never charge while driving. Check operation with a voltmeter: with the engine running both banks should read within 0.3 volts of each other; with the engine off and boost released they should read independently.

Entry step

The electric entry step, SKU STEP-3R, is a three-step unit that extends when the entry door opens and retracts when it closes. The step has its own 20 amp fuse in the chassis fuse block.

Retract logic

The step follows these rules. With the ignition off, the step extends when the door opens and retracts when the door closes. If the STEP OVERRIDE switch next to the door is on, the step stays extended while the door is closed, which is useful at a campsite. The step always retracts when the ignition is turned on, regardless of the override switch, and the dash STEP OUT light illuminates if it fails to retract. Do not drive with the STEP OUT light on. If the step will not retract, check the fuse, then check the door magnet switch on the door frame; a gap over 1/4 inch between magnet and switch causes the step to think the door is open. Lubricate the step pivots every 3 months with dry lubricant.

Water system

Model	Fresh water	Gray water	Black water
Summit 32	90 gal	65 gal	45 gal

The water heater is a THOR AquaLine WH-12, a 12 gallon gas and electric unit with the same direct-spark igniter (SKU WH-IGN-10) as the AquaLine WH-10. Winterizing follows the same steps as any THOR coach: drain the fresh tank and heater, bypass the heater, pump antifreeze through every fixture, and pour a cup into each drain trap. The Summit needs 4 gallons of antifreeze including the washing machine hookup.

Furnace and air conditioning

The furnace is a THOR ThermaCore TC-40 rated at 40,000 BTU, mounted in the basement compartment under the galley. It uses the same control board and sail switch (SKU FRN-SAIL) as the ThermaCore TC-30 and displays the same fault codes: F1 sail switch did not close, F2 high limit open, F3 ignition lockout after three attempts, F4 flame sensed with valve closed, F5 supply voltage below 10.5 volts. Reset by cycling the thermostat off for 5 minutes. Two 15,000 BTU roof air conditioners with heat pumps handle cooling and mild-weather heating; heat pump mode stops working below 40 degrees F, at which point the thermostat switches to the furnace.

Maintenance schedule

Item	Interval
Check tire pressure and DEF level	Before every trip
Test propane detector, smoke and CO alarms	Monthly
Run generator under load 30 minutes	Monthly
Clean and treat slide-out seals; lubricate entry step	Every 3 months
Inspect roof seams, reseal with ROOF-SEAL-Q as needed	Every 6 months
Check hydraulic reservoir level	Every 6 months
Engine oil, oil filter, fuel filter	12 months or 15,000 miles
Generator oil and filter	150 hours
Clean refrigerator burner and flue	12 months
Clean furnace burner and sail switch	12 months
Load test house and chassis batteries	12 months
Replace water heater anode	12 months
Torque lug nuts (450 ft-lb)	12 months
Hydraulic fluid change	60 months
Replace propane regulator	60 months

Troubleshooting

Symptom	Likely cause	Action
Chassis batteries dead after overnight stop	Isolator relay ISO-RLY-200 stuck closed	Use BATTERY BOOST to start; dealer to test and replace relay
House batteries not charging while driving	Isolator relay stuck open or 200 amp fuse blown	Check fuse at relay; dealer to test relay
Jacks creep down overnight	Air in hydraulic system	Perform bleed procedure; replace jack if creep continues
JACK DOWN light with jacks visibly up	Retract sensor misadjusted	Dealer to adjust sensor
Slide-out stalls partway	Low hydraulic fluid or cylinders out of sync	Check LOW FLUID light; cycle slide fully twice
Generator will not start	Fuel below 1/4 tank	Add fuel; check code on start panel
Refrigerator warm, ammonia smell at vent	Cooling unit leaked	Dealer to replace FRIDGE-CU

Refrigerator warm on propane, works on 120V	Dirty burner or flue	Clean burner; check propane pressure 11 inches water column
Entry step will not retract, STEP OUT light	Blown fuse or door magnet gap	Replace 20 amp fuse; adjust magnet to within 1/4 inch
Furnace code F1	Sail switch not closing	Charge batteries; clear return air; clean or replace FRN-SAIL
Furnace code F5	Voltage below 10.5	Charge house batteries; check battery terminals
Water stain at roof seam	Sealant failure	Reseal with ROOF-SEAL-Q; check for interior damage
Air brake alarm on startup	Normal below 60 psi	Wait for pressure to build above 100 psi before releasing brake